日频特征新增——落地

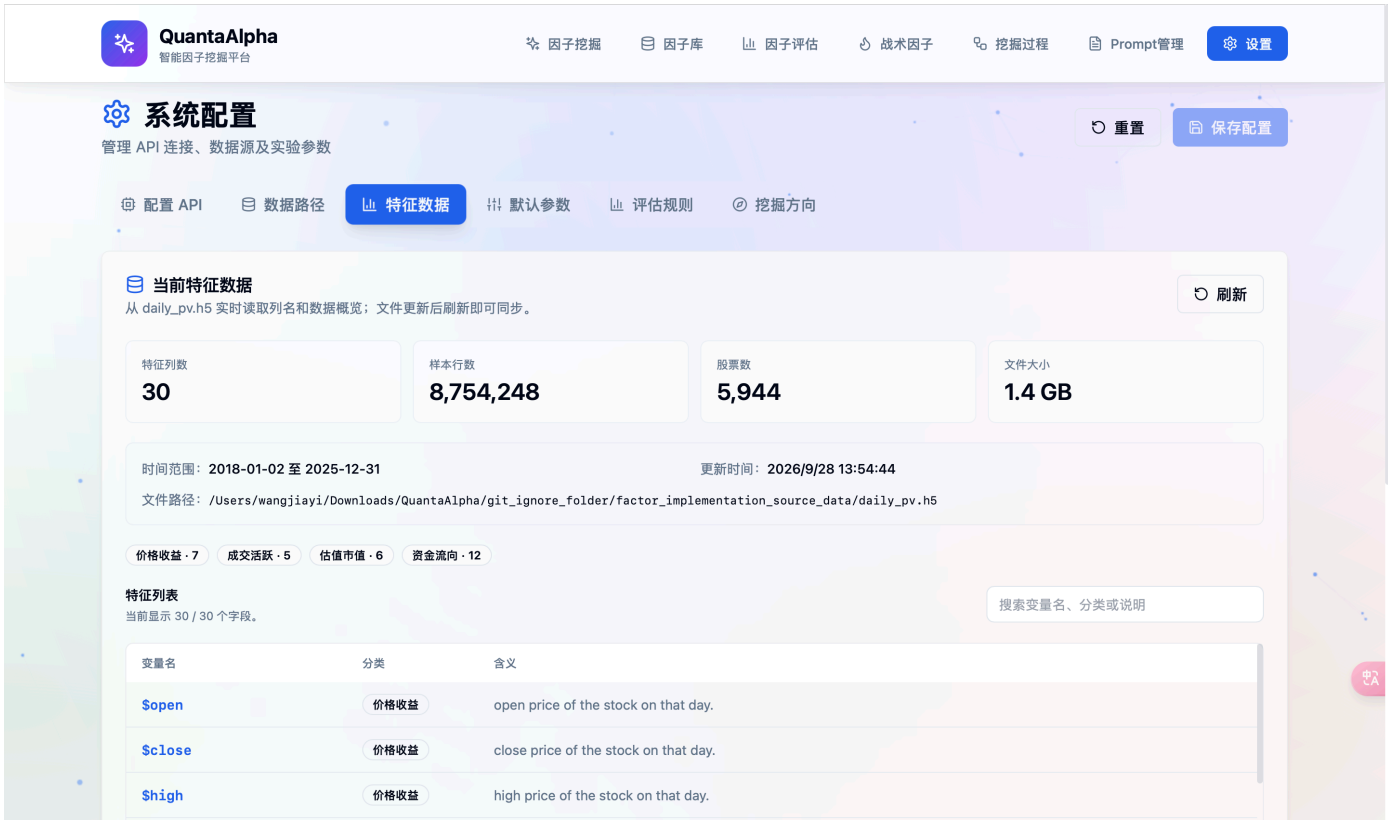
文件

新版 `daily_pv.h5` ：

- `/Users/wangjiayi/Downloads/QuantaAlpha/git_ignore_folder/factor_implementation_source_data/daily_pv.h5`
- 大小约 `1.4G`
- 形状： `8754248 x 30`
- 时间范围： `2018-01-02` 到 `2025-12-31`
- 股票数： `5944`

前端新增特征数据查看页面

首页-设置-特征数据：



测试

测试问题：

```
挖掘估值和主动买卖力量结合的日频因子，优先使用 pe、pe_ttm、buying、selling、小单和大单资金流字段
```

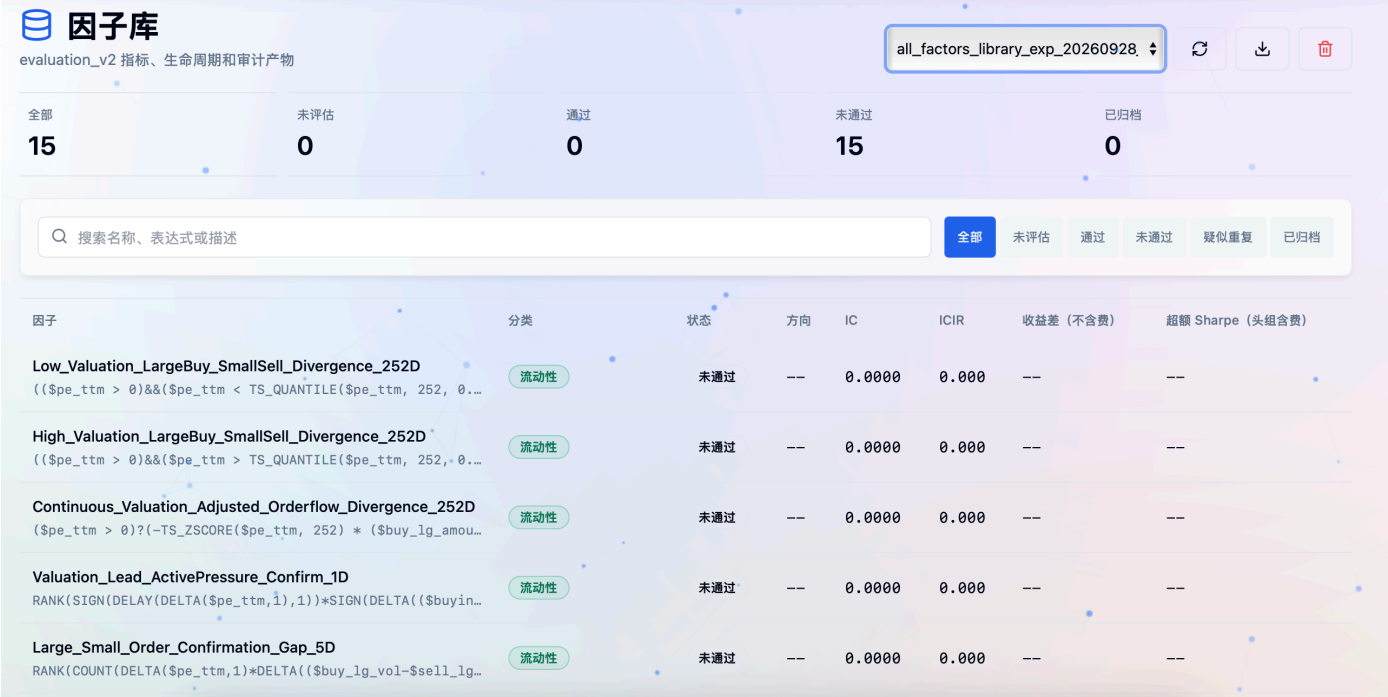
检查：
1.跑完后看生成公式里是否出现这些新变量：比如：

- $pe
- $pe_ttm
- $buying
- $selling
- $buy_sm_amount
- $sell_sm_amount
- $buy_lg_amount
- $sell_lg_amount

2.点开因子详情，看是否能正常回测、有无 `FactorEmptyError` 或字段不存在报错。

第一次：

跑出了因子，但没办法回测：



出现的errors：

```
2026-09-28 14:24:43.911 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败：变量替换时发生子串污染，导致表达式中的 pe_ttm、buy_lg_amount 等变量名被错误替换并产生 SyntaxError，未生成 result.h5，因子值缺失，因此实现不正确。"}[0m

2026-09-28 14:24:57.900 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Low_Valuation_LargeBuy_SmallSell_Divergence_252D: No factor values generated for Low_Valuation_LargeBuy_SmallSell_Divergence_252D

14:24:57❌2026-09-28 14:24:57.923 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

```

```
14:24:57❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:24:57❌2026-09-28 14:24:57.943 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for High_Valuation_LargeBuy_SmallSell_Divergence_252D: No factor values generated for High_Valuation_LargeBuy_SmallSell_Divergence_252D

14:24:57❌2026-09-28 14:24:57.963 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:24:57❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for High_Valuation_LargeBuy_SmallSell_Divergence_252D

14:24:57❌2026-09-28 14:24:57.974 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Continuous_Valuation_Adjusted_Orderflow_Divergence_252D: No factor values generated for Continuous_Valuation_Adjusted_Orderflow_Divergence_252D

14:24:57❌2026-09-28 14:24:57.993 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

❌2026-09-28 14:26:27.850 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败：变量替换逻辑存在前缀冲突，`$pe_ttm` 被错误替换为 `df['$pe']_ttm`，导致语法错误并抛出 SyntaxError，未能生成 result.h5。因子语义本身（昨日 pe_ttm 变化方向与今日主动买卖压力变化方向同向确认后截面排序）与定义一致，问题出在实现层面。建议在字符串... [270 chars][0m

❌2026-09-28 14:26:37.915 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败：变量替换将 $pe_ttm 错误替换为 df['$pe']_ttm，导致 SyntaxError 且未生成 result.h5；需修复 $pe 与 $pe_ttm 等前缀冲突后重新计算因子。"}[0m

2026-09-28 14:26:54.991 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Valuation_Lead_ActivePressure_Confirm_1D: No factor values generated for Valuation_Lead_ActivePressure_Confirm_1D

14:26:55❌2026-09-28 14:26:55.002 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

2026-09-28 14:26:55.009 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Large_Small_Order_Confirmation_Gap_5D: No factor values generated for Large_Small_Order_Confirmation_Gap_5D

14:26:55❌2026-09-28 14:26:55.015 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:26:55❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:26:55❌2026-09-28 14:26:55.023 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Pressure_Lead_Valuation_Confirm_Decay_10D: No factor values generated for Pressure_Lead_Valuation_Confirm_Decay_10D
```

```
14:26:55❌2026-09-28 14:26:55.031 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:26:55❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for Pressure_Lead_Valuation_Confirm_Decay_10D

2026-09-28 14:30:29.679 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "因子代码执行失败且未生成 result.h5，原因是变量替换存在子串冲突：amount 被错误替换到 buy_lg_amount、sell_lg_amount 等变量内部，导致 NameError: name 'buy_lg_df' is not defined；需改为按完整 token 或按变量名长度从长到... [209 chars][0m

2026-09-28 14:30:48.560 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行时因 NameError: name 'buy_lg_df' is not defined 中断，未生成 result.h5，且大单买卖金额列名或模板变量替换有误，故实现不正确。"}[0m

2026-09-28 14:30:58.855 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败，因变量替换错误导致 NameError: name 'buy_lg_df' is not defined，未生成 result.h5；需修正变量到原始数据列的映射（如 buy_lg_amount 等）后重新运行。"}[0m2026-09-28 14:30:58.915 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Hierarchy_Divergence_CloseRank_20D: No factor values generated for Hierarchy_Divergence_CloseRank_20D

14:30:58❌2026-09-28 14:30:58.921 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:30:58❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for Hierarchy_Divergence_CloseRank_20D

14:30:58❌2026-09-28 14:30:58.927 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Active_Imbalance_Large_TrendSign_5D: No factor values generated for Active_Imbalance_Large_TrendSign_5D

14:30:58❌2026-09-28 14:30:58.933 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

2026-09-28 14:30:58.939 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Large_Buy_Small_Sell_Breakout_20D: No factor values generated for Large_Buy_Small_Sell_Breakout_20D

14:30:58❌2026-09-28 14:30:58.944 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:30:58❌raise FactorEmptyError(f"No factor values generated for {factor_name}")
```

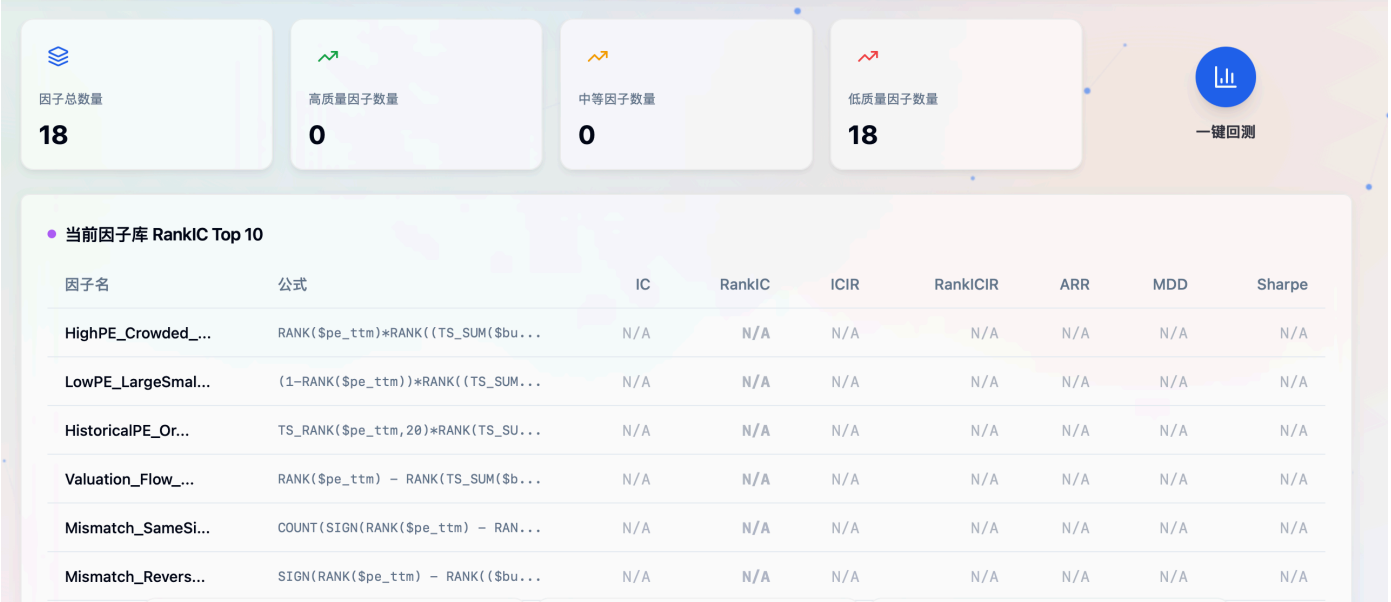
```
❌2026-09-28 14:33:24.692 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "实现未通过：变量替换时 $pe_ttm 被错误替换为 df['$pe']_ttm，导致表达式 SyntaxError，代码未成功执行且未生成 result.h5 因子值；应按完整变量名精确替换或按变量名长度降序替换后重跑。"}[0m

2026-09-28 14:33:51.875 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败，变量替换阶段将 $pe_ttm、$buy_sm_amount 等子串误替换，导致 SyntaxError 且未生成 result.h5；需按完整 token/最长列名匹配替换后重新运行。"}[0m
```

进行了以下修复：

第二次：

仍然跑出了因子，但没办法回测：



出现的errors：

2026-10-12 14:06:44.411 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败，pe_ttm被错误替换为$df['$pe']ttm，触发 SyntaxError 且未生成 result.h5；同时未严格实现“大单和小单净买入均为正时才取正向资金流强度”的条件，因此实现不正确。"}[0m❌2026-10-12 14:07:03.758 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "执行失败：变量替换存在前缀冲突，pe先于pe_ttm 被替换，导致生成 df['$pe']ttm 并触发 SyntaxError，未生成 result.h5，因此因子实现不正确。"}[0m

14:07:03•2026-10-12 14:07:03.770 | INFO | quantaalpha.coder.costeer.evaluators:evaluate:109 - Final decisions: [False, False, False] (0/3 passed)

14:07:03•2026-10-12 14:07:03.791 | INFO | quantaalpha.log.time:wrapper:16 - factor_calculate took 37.73s

14:07:03•Workflow Progress: 40%|██████ | 2/5 [01:00<00:34, 11.66s/step, loop_index=0, step_index=2, step_name=factor_calculate]Workflow Progress: 60%|████████ | 3/5 [01:00<00:47, 23.57s/step, loop_index=0, step_index=2, step_name=factor_calculate]2026-10-12 14:07:03.800 | INFO | quantaalpha.pipeline.loop:factor_backtest:199 - Start factor backtest (Local: True)

14:07:03❌2026-10-12 14:07:03.811 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for HighPE_Crowded_Buy_Alignment_5D: No factor values generated for HighPE_Crowded_Buy_Alignment_5D

14:07:03❌2026-10-12 14:07:03.817 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:07:03•File "/Users/wangjiayi/Downloads/QuantaAlpha/quantaalpha/evaluation/runner.py", line 50, in develop

14:07:03❌2026-10-12 14:07:03.824 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for LowPE_LargeSmall_Divergence_10D: No factor values generated for LowPE_LargeSmall_Divergence_10D

14:07:03❌2026-10-12 14:07:03.830 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:07:03❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:07:03❌2026-10-12 14:07:03.835 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for HistoricalPE_OrderFlow_Interaction_20D: No factor values generated for HistoricalPE_OrderFlow_Interaction_20D

14:07:03❌2026-10-12 14:07:03.841 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:07:03❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for HistoricalPE_OrderFlow_Interaction_20D

❌2026-10-12 14:08:21.048 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "代码执行失败：变量替换时因按子串替换导致 pe误匹配pe_ttm、amount 误匹配 buy_lg_amount/sell_lg_amount，生成了非法表达式并触发 SyntaxError，未生成 result.h5，因此因子实现不正确；应改为按完整变量名或按长度降序替换。"}[0m

❌2026-10-12 14:08:33.267 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "代码执行失败并抛出 SyntaxError，未生成 result.h5；原因是 pe_ttm、buy_lg_amount、$sell_lg_amount 等下划线变量被错误解析为非法表达式，同时 COUNT(SIGN(M_t)*SIGN(DELAY(M,1))>0,5) 统计的是相邻两日同向天数，并非描述... [234 chars][0m ❌2026-10-12 14:08:43.309 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败：AST校验虽通过，但因变量名前缀重叠导致模板替换出错，`$pe_ttm` 被错误替换为 `df['$pe']_ttm`，buy_lg_amount\sell_lg_amount被错拆为buy_lg_df['$amount']` 等，产生 SyntaxError 且未生成 res... [240 chars][0m

❌2026-10-12 14:08:43.386 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Valuation_Flow_Mismatch_5D: No factor values generated for Valuation_Flow_Mismatch_5D

14:08:43❌2026-10-12 14:08:43.395 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:08:43•File "/Users/wangjiayi/Downloads/QuantaAlpha/quantaalpha/evaluation/runner.py", line 50, in develop

14:08:43❌2026-10-12 14:08:43.405 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Mismatch_SameSign_Days_5D: No factor values generated for Mismatch_SameSign_Days_5D

14:08:43❌2026-10-12 14:08:43.416 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:08:43❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:08:43❌2026-10-12 14:08:43.431 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Mismatch_Reversal_From_20D_Extreme: No factor values generated for Mismatch_Reversal_From_20D_Extreme

14:08:43❌2026-10-12 14:08:43.445 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:08:43❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for Mismatch_Reversal_From_20D_Extreme

❌2026-10-12 14:16:22.059 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Large_Order_Mismatch_PE_TTM_State: No factor values generated for Large_Order_Mismatch_PE_TTM_State

14:16:22❌2026-10-12 14:16:22.065 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:16:22❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

❌2026-10-12 14:17:44.814 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "代码执行失败：表达式变量替换错误，导致 buy_lg_amount 等列名被错误替换为未定义变量（如 buy_lg_df），触发 NameError，且未生成 result.h5，因子值未能计算。"}[0m

❌2026-10-12 14:17:53.098 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "代码执行失败：表达式变量替换后引用未定义的buy_lg_df，导致NameError且未生成result.h5；需修复解析/替换逻辑，确保正确引用df中的buy_lg_amount、sell_lg_amount、buy_sm_amount、sell_sm_amount列并重新执行。"}[0m

❌2026-10-12 14:18:07.847 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "执行失败：变量替换存在子串冲突，替换 amount 时误改了 buy_lg_amount 和 sell_lg_amount，导致 NameError: name 'buy_lg_df' is not defined，且未生成 result.h5；需按完整变量名匹配或优先替换较长变量名后重跑。"}[0m

❌2026-10-12 14:18:07.899 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Institution_Retail_Divergence_Mean_5D: No factor values generated for Institution_Retail_Divergence_Mean_5D

14:18:07❌2026-10-12 14:18:07.906 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:18:07❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:18:07❌2026-10-12 14:18:07.913 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Opposite_Flow_Same_Sign_Count_5D: No factor values generated for Opposite_Flow_Same_Sign_Count_5D

14:18:07❌2026-10-12 14:18:07.920 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:18:07❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for Opposite_Flow_Same_Sign_Count_5D

14:18:07❌2026-10-12 14:18:07.926 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Large_Flow_Price_Absorption_Residual_5D: No factor values generated for Large_Flow_Price_Absorption_Residual_5D

14:18:07❌2026-10-12 14:18:07.932 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

❌2026-10-12 14:20:03.637 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "实现失败：变量替换逻辑错误导致表达式出现非法语法（如 df['$pe']_ttm），执行抛出 SyntaxError 且未生成 result.h5，无法得到因子值，因此判定不正确。"}[0m

❌2026-10-12 14:20:13.120 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse: {"final_decision": false, "final_feedback": "代码执行失败且未生成 result.h5；变量替换时将 pe_ttm、buy_lg_amount 等完整列名错误拆分，导致 SyntaxError。虽然因子表达式语义与定义基本一致，但实现存在严重错误，需修复完整变量名匹配/替换逻辑后重新运行。"}[0m

❌2026-10-12 14:20:23.233 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for HighVal_SameDirection_Buy_Count_5D: No factor values generated for HighVal_SameDirection_Buy_Count_5D

14:20:23❌2026-10-12 14:20:23.238 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:20:23❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for HighVal_SameDirection_Buy_Count_5D

14:20:23❌2026-10-12 14:20:23.244 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for LowVal_LargeBuy_SmallSell_Persistence_5D: No factor values generated for LowVal_LargeBuy_SmallSell_Persistence_5D

14:20:23❌2026-10-12 14:20:23.250 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:20:23•File "/Users/wangjiayi/Downloads/QuantaAlpha/quantaalpha/evaluation/runner.py", line 50, in develop

14:20:23❌2026-10-12 14:20:23.256 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for HighVal_LargeInflow_LowAbsorption_5D: No factor values generated for HighVal_LargeInflow_LowAbsorption_5D

14:20:23❌2026-10-12 14:20:23.262 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:20:23❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

❌2026-10-12 14:21:53.857 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:comment 1: 表达式语义与因子定义基本一致，但变量命名触发了模板中的字符串替换冲突：`$pe_ttm` 被错误替换成 `df['$pe']_ttm`，`$buy_lg_amount/$sell_lg_amount` 也被拆成了 `buy_lg_df['$amount']` 等非法形式，导致 SyntaxError。这是表达式层面的关键可执行性问题，需避免使用会被较短列名子串误替换的变量形式... [311 chars][0m

❌2026-10-12 14:21:56.975 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败：变量名替换冲突导致 pe_ttm被错误替换为$df['$pe']_ttm、$buy_lg_amount/$sell_lg_amount 被拆成 buy_lg_df['$amount'] 等非法形式，eval 时抛出 SyntaxError，未生成 result.h5，无任何因子值输出，故实现... [280 chars][0m

❌2026-10-12 14:22:32.883 | INFO | quantaalpha.llm.client:collect_response:1072 - [96mResponse:{"final_decision": false, "final_feedback": "代码执行失败，未生成 result.h5：变量替换环节存在前缀误匹配问题，pe_ttm被短变量pe 部分替换成 df['$pe']_ttm，buy_lg_amount/$buy_sm_amount$中的amount 也被单独替换，导致 eval 时抛出 SyntaxError。因子逻辑本身与定义... [579 chars][0m

❌2026-10-12 14:22:32.944 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Value_Flow_Mismatch_Convergence_20D: No factor values generated for Value_Flow_Mismatch_Convergence_20D

14:22:32❌2026-10-12 14:22:32.951 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

❌2026-10-12 14:22:32.958 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Large_Small_Order_Divergence_Value_10D: No factor values generated for Large_Small_Order_Divergence_Value_10D

14:22:32❌2026-10-12 14:22:32.964 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:22:32❌raise FactorEmptyError(f"No factor values generated for {factor_name}")

14:22:32❌2026-10-12 14:22:32.971 | ERROR | quantaalpha.evaluation.runner:develop:73 - OTO evaluation failed for Crowded_Buying_Reversal_Risk_5D: No factor values generated for Crowded_Buying_Reversal_Risk_5D

14:22:32❌2026-10-12 14:22:32.977 | ERROR | quantaalpha.evaluation.runner:develop:74 - Traceback (most recent call last):

14:22:32❌quantaalpha.core.exception.FactorEmptyError: No factor values generated for Crowded_Buying_Reversal_Risk_5D

出现的errors及原因：

1. **`SyntaxError`：变量名前缀冲突**
   `$pe_ttm` 被错误替换成 `df['$pe']_ttm`，表达式无法执行。
2. **`NameError: buy_lg_df is not defined`：金额字段被拆坏**
   替换 `amount` 时误改了 `buy_lg_amount` 等字段，变成 `buy_lg_df['$amount']`。
3. **`FactorEmptyError / OTO evaluation failed`：没有因子值**
   上面的错误导致未生成 `result.h5`，回测拿不到因子值。页面显示因子名称和公式，不代表已经计算成功。

原因：第一次修复已覆盖部分回测计算入口，但挖掘用的 `template.jinjia2` 仍使用旧的字符串替换逻辑，所以第二次继续报同样的错。

另外，日志还提示个别公式漏了“大单和小单净买入均为正”的条件，以及 `COUNT` 统计口径与描述不一致，需单独核对。

第二次修复：

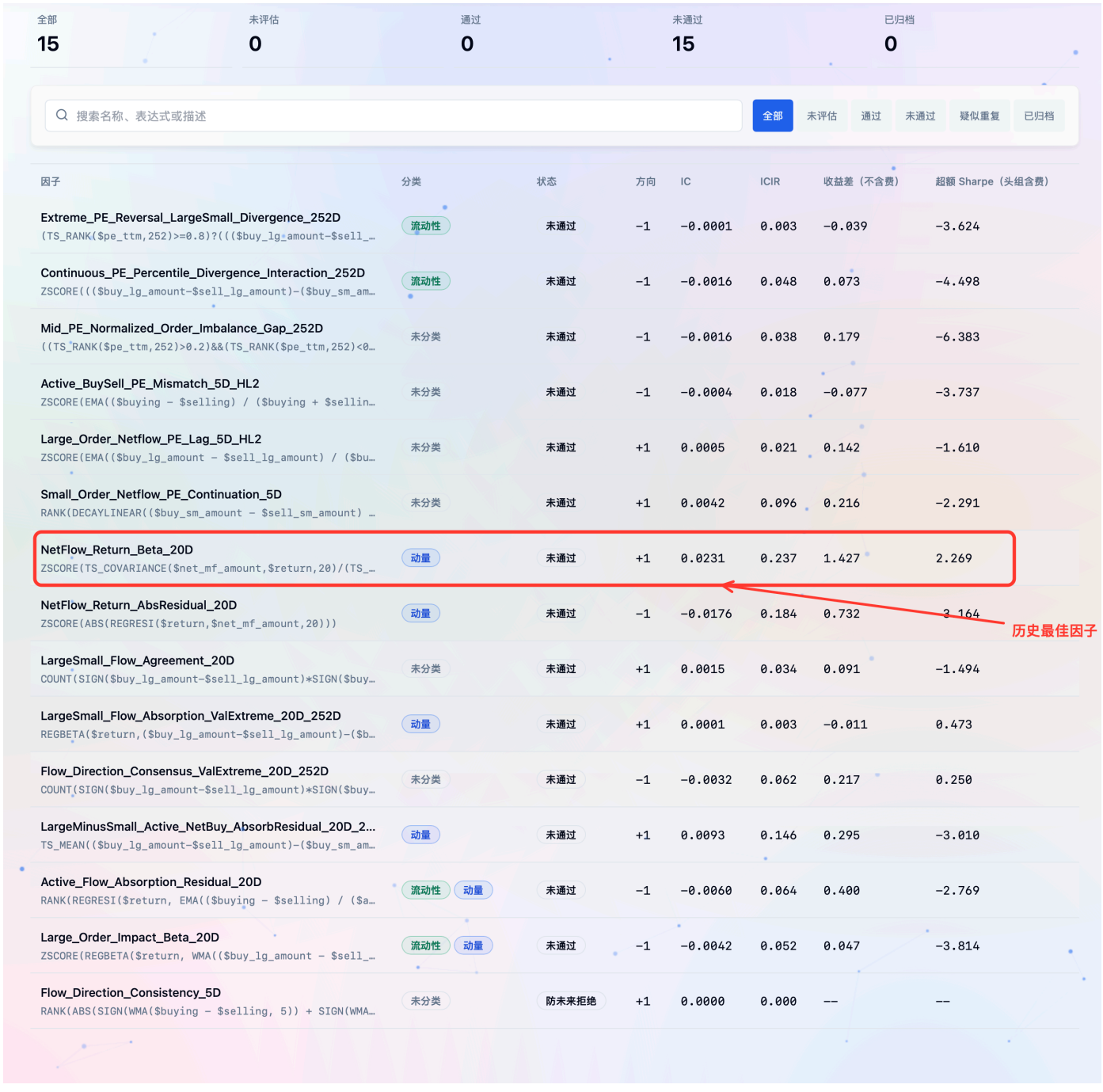
- 挖掘模板改用 `replace_data_column_tokens`，按完整字段名替换，避免 `pe/pe_ttm`、`amount/大小单金额` 相互影响。
- 新增模板执行测试，覆盖估值、大小单资金流、主动买卖字段，以及字段顺序变化。
- 已重启挖掘后端，健康检查正常。

修复结果：

- ✅ 5 项相关测试全部通过。
- ✅ 真实数据末尾抽样 10,000 行，成功生成 `result.h5`，得到 9,711 条非空因子值，与 pandas 直接计算结果一致。
- ✅ 上述字段替换错误在验证中未再出现。
- 待验证：新建任务重跑完整挖掘和回测；旧任务中的错误脚本不会自动更新，公式语义问题也尚未逐个修复。

第三次:

成功运行完成了，运行出了15个因子，有一个没有结果，原因是防未来拒绝，待查明，一个因子表现不错，为历史最佳因子：



因子	分类	状态	方向	IC	ICIR	收益差（不含费）	超额 Sharpe（头组含费）
Extreme_PE_Reversal_LargeSmall_Divergence_252D (TS_RANK($pe_ttm,252)>=0.8)?(($buy_lg_amount-$sell_…	流动性	未通过	-1	-0.0001	0.003	-0.039	-3.624
Continuous_PE_Percentile_Divergence_Interaction_252D ZSCORE((($buy_lg_amount-$sell_lg_amount)-($buy_sm_am…	流动性	未通过	-1	-0.0016	0.048	0.073	-4.498
Mid_PE_Normalized_Order_Imbalance_Gap_252D ((TS_RANK($pe_ttm,252)>0.2)&&(TS_RANK($pe_ttm,252)<0…	未分类	未通过	-1	-0.0016	0.038	0.179	-6.383
Active_BuySell_PE_Mismatch_5D_HL2 ZSCORE(EMA(($buying - $selling) / ($buying + $sellin…	未分类	未通过	-1	-0.0004	0.018	-0.077	-3.737
Large_Order_Netflow_PE_Lag_5D_HL2 ZSCORE(EMA(($buy_lg_amount - $sell_lg_amount) / ($bu…	未分类	未通过	+1	0.0005	0.021	0.142	-1.610
Small_Order_Netflow_PE_Continuation_5D RANK(DECAYLINEAR(($buy_sm_amount - $sell_sm_amount) …	未分类	未通过	+1	0.0042	0.096	0.216	-2.291
NetFlow_Return_Beta_20D ZSCORE(TS_COVARIANCE($net_mf_amount,$return,20)/(TS_…	动量	未通过	+1	0.0231	0.237	1.427	2.269
NetFlow_Return_AbsResidual_20D ZSCORE(ABS(REGRESI($return,$net_mf_amount,20)))	动量	未通过	-1	-0.0176	0.184	0.732	3.164
LargeSmall_Flow_Agreement_20D COUNT(SIGN($buy_lg_amount-$sell_lg_amount)*SIGN($buy…	未分类	未通过	+1	0.0015	0.034	0.091	-1.494
LargeSmall_Flow_Absorption_ValExtreme_20D_252D REGBETA($return,($buy_lg_amount-$sell_lg_amount)-($b…	动量	未通过	+1	0.0001	0.003	-0.011	0.473
Flow_Direction_Consensus_ValExtreme_20D_252D COUNT(SIGN($buy_lg_amount-$sell_lg_amount)*SIGN($buy…	未分类	未通过	-1	-0.0032	0.062	0.217	0.250
LargeMinusSmall_Active_NetBuy_AbsorbResidual_20D_2... TS_MEAN(($buy_lg_amount-$sell_lg_amount)-($buy_sm_am…	动量	未通过	+1	0.0093	0.146	0.295	-3.010
Active_Flow_Absorption_Residual_20D RANK(REGRESI($return, EMA(($buying - $selling) / ($a…	流动性 动量	未通过	-1	-0.0060	0.064	0.400	-2.769
Large_Order_Impact_Beta_20D ZSCORE(REGBETA($return, WMA(($buy_lg_amount - $sell_…	流动性 动量	未通过	-1	-0.0042	0.052	0.047	-3.814
Flow_Direction_Consistency_5D RANK(ABS(SIGN(WMA($buying - $selling, 5)) + SIGN(WMA…	未分类	防未来拒绝	+1	0.0000	0.000	--	--